

What's New — SubDocentAI

Changes are listed newest-first. Dates are commit dates.

2026-03-10 — FAQ Corpus: pam_001–pam_226 (226 entries)

The custom Pampanito FAQ corpus grew from 7 seed entries to 226 in a series of batches, each tested with a regression suite before committing.

Batch	Range	Count	Topics added
Seed	pam_001–007	7	Maneuvering room, bunks in torpedo rooms, engines, food storage, smarts, torpedo reloads, running submerged
1–2	pam_008–034	27	Navigation, sonar, radio, Pampanito history, crew life basics, weapons, attack procedure, visiting
3	pam_035–066	32	Crew size, recreation, toilets, surfacing, periscopes, conning tower, depth control, losses, torpedoes, captain, Silent Service, crew demographics, resupply, smell, mail, return to port, wolf pack, POW rescue, KIA, Japanese subs, US vs U-boats
4	pam_067–087	21	Torpedo count, temperature, noise, patrol length, enemy sighting, sub-vs-sub combat, bottoming, largest sub, war patrol, special forces, laundry, shore leave, ports, Mark 14, crew tracking, battery room, crew after war, medals, fuel, diving alarm
5	pam_088–112	25	Food, submarine tender, SS designation, POW identification, doctors, submarine school, SJ radar, war strategy, patrol assignments, damage, end-of-war, commissioning/decommissioning, museum preservation, longest submerged
6	pam_113–132	20	Lifeguard mission, mine laying, depth charge sensation, hot bunking, deck gun operation, Pampanito name, fire fighting, snorkel, Battle of Midway, radio direction finding, claustrophobia, galley, fresh water, close calls, convoy attack, seasickness, warship sinkings, aircraft threat, wolfpack tactics, spies/special forces

Batch	Range	Count	Topics added
7-10	pam_133-197	65	Battle stations, pharmacist's mate, ballast tanks, enemy minefields, fleet boats, battery recharging, Gato vs Balao, captain's duties, destroyer evasion, torpedo sounds, commissioning day, Japanese sub fate, crew death, patrol orders, control room, forward torpedo room, garbage disposal, enemy survivors, flooding/fire, periscope feather, boredom, TDC operator, grounding, Japanese convoys, battle stars, flying bridge, Pampanito dive depth, magnetic exploder, top commander, celebration, return to port, Silent Service origin, construction changes, snorkel, periscope depth, OOD duties, conning tower, crew training, sound-powered phones, night identification, after torpedo room, Navy selection, crew of sunken sub, night surface attacks, lookouts, Pacific refueling, angle on bow, Pearl Harbor comms, forward engine room, navigation to patrol area, surfacing decision
11	pam_198-210	13	Escape trunk, sonar tracking, chief of the boat, torpedo spreads, collision avoidance, periscope attack procedure, after engine room, diving planes, WWII radar use, tropical heat, commander qualities, depth charge interior, destroyer evasion tactics
12	pam_211-226	16	Submarine tender services, Mark 14 depth failure, ULTRA codebreaking, down-the-throat shot, mine threats, sinking accuracy, strategic impact, torpedo tube mechanics, forward battery/officers' quarters, Medal of Honor, crew mail, deck gun vs torpedoes, conning tower vs control room, Presidential Unit Citation, two-periscope design, Pampanito war record

All 226 entries pass regression tests across all 12 batches.

2026-03-10 — Scoring Engine Fixes

Several retrieval engine bugs were fixed during batch 12 development:

- **wants_mark_compare over-trigger fixed** — previously fired for any torpedo query containing "mark", inflating [faq_982](#) (encyclopedic torpedo comparison FAQ) to score 36.0 on unrelated queries. Narrowed to require both Mark 14 and Mark 18 in query, or explicit comparison vocabulary (versus, difference, compare, etc.).
- **raw_lower variable order bug fixed** — [raw_lower](#) was referenced before assignment in [detect_intent\(\)](#); moved definition before the [wants_mark_compare](#) block.
- **Prepositions added to STOPWORDS** — "into", "onto", "upon", "within", "without", "through", "throughout" now filtered from query tokens, preventing them from blocking [all_q_covered](#) title-match checks.
- **fired synonym entry added** — expands to launch, launched, fire, shoot, shot, torpedo, tube; allows "how was a torpedo fired" to match FAQ titles using "launch".
- **target synonym expanded** — added "targets" so updated FAQ titles containing the plural form score correctly.

- **Title-bonus formula corrected** — `effective_weight = max(weight, weight * 4.0 * coverage)` ensures a perfect-coverage title always beats a partial one regardless of base weight.
-

2026-03-04-05 — Android Speech Recognition

Full voice input support added for Android browsers, which cannot use the browser's native Web Speech API against a local HTTPS server.

- **/transcribe POST endpoint** — accepts an audio file (WebM/OGG from MediaRecorder), sends to Groq `whisper-large-v3-turbo`, returns transcribed text (~0.3 s latency).
 - **/health response extended** — now includes `transcribe_available: true/false` so the client can decide which speech path to use on startup.
 - **Client dual-path routing** — on load, client calls `/health`; if `transcribe_available`, all voice input routes through the record → Groq path (bypassing native SpeechRecognition even on Android Chrome, which has `webkitSpeechRecognition` but fails with "No speech detected" on local servers).
 - **Text input fallback** — on browsers with no microphone support at all (Android Firefox, Samsung Internet), the mic button converts to a keyboard icon and a text input row appears immediately.
 - **Groq replaces OpenAI Whisper** — same `openai` Python client, different `base_url` and `GRQ_API_KEY`; faster and cheaper.
-

2026-03-04 — UX Improvements

- Password gate auto-submits on entering the 4th digit (no Enter button needed).
 - FAQ source ID displayed below answer text when answer comes from the FAQ corpus.
 - Race condition eliminated in speech initialisation — `enableRecordMode()` called synchronously.
 - Default `transcribeAvailable=true` so TLS cert failures on first load don't silently break speech.
-

2026-03-02-03 — Retrieval Quality & Compartment Awareness

- **Compartment-aware retrieval** — query is matched against the visitor's current tour stop; tour chunks in that compartment score higher than off-compartment chunks.
 - **Tour stop attribution** — answers from the audio tour corpus are prefixed with "From the audio tour in [location]."
 - **Intent detection** — WHERE questions prepend "In the [location]." to the answer; HOW MANY questions activate a quantity-word score multiplier.
 - **STT corrections** — phonetic substitutions added for common misrecognitions (e.g., "banks" → "bunks", torpedo-room homophones).
 - **Synonym expansion** — domain synonyms added in batches alongside each new FAQ group (e.g., communicate, torpedo, fast/faster, crew, radar, medical, museum, storage, snorkel, wolfpack, convoy).
 - **STOPWORDS tuning** — removed "happened/happen" (too domain-specific to suppress); added directional and universal submarine terms.
-

2026-03-02 — Feedback System

- Thumbs up / thumbs down widget on each answer.
 - Optional free-text comment field.
 - Responses logged to `feedback.jsonl`.
 - `GET /feedback/list` endpoint returns recent entries.
 - Password-gated feedback viewer page (`feedback.html`).
-

2026-03-01-02 — Infrastructure & Launch

- Deployed to Render.com (`subdocentai.com`) with `render.yaml`.
- Added `GROQ_API_KEY` to Render environment for production Whisper transcription.
- Renamed `tour.html` → `pampanito.html`; added `/pampanito.html` redirect and root redirect to index.
- Password gate added to protect pre-launch preview.
- "Coming soon" index page for public-facing URL.
- Self-signed TLS certificates for local HTTPS (required for microphone access).
- `start_https.sh` — single-command server start/restart.